

Statistics 250, Section 701, Quiz 3 (75 Points)

June 10, 1998, Dr. Jürgen Symanzik

Your Name: _____

Question 1: Short Answers (17 Points)

1. List a sample space Ω_Q that shows the different ways in which three true-false questions can be answered. (2 Points)

2. Assume that students are only guessing when answering the three questions in (1) above, i. e., each sample point from Ω_Q is equally likely to occur. Let Q2 be the event that question 2 is answered correctly and let Q3 be the event that question 3 is answered correctly. Indicate

$$P(Q2) =$$

$$P(Q3) =$$

$$P(Q2 \text{ and } Q3) =$$

Based on your answers above, are the events Q2 and Q3 dependent or independent. Explain why. (6 Points)

3. A fast food restaurant has three employees working in the kitchen. Employee A prepares 60 percent, employee B 30 percent, and employee C 10 percent of the french fries. Of the french fries prepared by employees A, B, and C, 1%, 2%, and 5%, respectively, contain no salt. Find the probability $P(U)$ that you obtain unsalted french fries:

$$P(U) =$$

Now assume you got unsalted french fries. What is the probability that they have been prepared by employee B? (**5 Points**)

4. The Swiss lottery has a game called “7 out of 39”. You make a selection of 7 numbers between 1 and 39 and you win the big prize if exactly these numbers are drawn in the weekly drawing. How many different combinations are possible to select 7 out of 39 numbers? Calculate this value!

Compare this with the German lottery game “6 out of 49” which has 13,983,816 possible combinations. In which country do you have higher chances to win the big prize? (**4 Points**)

Question 2: Simple Probabilities (**11 Points**)

Recall the experiment from class where students have been asked on their expected performance in the second quiz. Students had to assign numbers 1 (for expected best performance) through 4 (for expected worst performance) to Questions 1 through 4. Assume that this assignment of numbers to questions is made randomly. A reasonable sample space Ω_C that describes this experiment when considering the outcome of one single student only is

$$\begin{aligned}\Omega_C = \{ & 1234, 1243, 1423, 1432, 1324, 1342, \\ & 2134, 2143, 2413, 2431, 2314, 2341, \\ & 3214, 3241, 3421, 3412, 3124, 3142, \\ & 4231, 4213, 4123, 4132, 4321, 4312\}\end{aligned}$$

1. We assume that each sample point from Ω_C is equally likely to occur. Indicate (**5 Points**)

$$|\Omega_C| =$$

$$P(\{1342\}) =$$

$$P(1 \text{ occurs in 3rd position}) =$$

$$P(2 \text{ occurs in 4th position}) =$$

$$P(1 \text{ occurs in 3rd position and } 2 \text{ occurs in 4th position}) =$$

2. Based on your answers in (1) above, are the events “1 occurs in 3rd position” and “2 occurs in 4th position” dependent or independent. Explain why. (**6 Points**)

Question 3: Dependence and Independence (**16 Points**)

The following table indicates the final grades of Stat 250 students split by university in a previous year:

	GMU	not GMU	Σ
A		7	15
B	4	8	
C	2		
D	1	3	4
Σ	15	20	

1. Fill in the missing 5 numbers in the table above. (**5 Points**)
2. Indicate (**4 Points**)

$$P(\text{Grade A})$$

$$P(\text{not GMU})$$

$$P(\text{Grade B} \mid \text{GMU})$$

$$P(\text{Grade A and not GMU})$$

3. Explain how you can calculate $P(\text{Grade D and GMU})$ based on $P(\text{Grade D})$ and $P(\text{GMU} \mid \text{Grade D})$ and **formally** obtain the result: (**4 Points**)

4. Are “Grade” and “University” independent or dependent? Explain why? (**3 Points**)

Question 4: Let's Make a Deal #2 (10 Points)

During the last year, several Stat 250 students participated at the “Let's Make a Deal” game show and won big money for a considerable amount of times (about 66%). Based on this fact, the showmaster decided to make the game a little bit tougher and have 5 doors of which 2 are opened after the player's first guess. The player can stay with the first choice or select another one (BUT NOT BOTH!) of the remaining two unopened doors.

Formally calculate the probability of the events

B1: “Player switches and wins” and

B2: “Player does not switch and wins”.

It might be helpful to start with making reasonable assumptions for the following events (assuming a “fair” game)

A1: “First selected door contains money”

A2: “First selected door does not contain money”

and make use of known conditional probabilities related to these two events A1 and A2. Diagrams might be helpful. So, which strategy is best in this 5 door “Let's Make a Deal” game?

Question 5: WebStat 1.0 (21 Points)

The following questions are related to the features of WebStat 1.0 which is available at <http://www.stat.sc.edu/~west/webstat/version1.0/>

1. Name **two** of the plot types we know from class that are accessible through the “Graphics” menu in WebStat 1.0. **(2 Points)**
2. We calculated five different summary statistics over and over again for our data sets used in class. All these summary statistics can also be calculated through the WebStat 1.0 “Stat” and “Summary Stat” menu. Name all **five** of them. **(2.5 Points)**
3. WebStat 1.0 calculates the “Variance”. Is this the **sample variance** or **population variance**? **(2 Points)**
4. The following stem-and-leaf plot has been created using WebStat 1.0. It displays the number of wins in the “Cost of Victories” example. Does it appear to be correct? Either answer **yes** or answer **no** and list **all** errors and differences with respect to stem-and-leaf plots as discussed in class.

Variable: wins

```
1 : 144
1 : 556667789999999
2 : 0000124444
2 : 6
High: 29.0
```

The original data is 8, 20, 17, 17, 20, 19, 19, 19, 14, 19, 15, 24, 22, 15, 29, 24, 26, 21, 24, 20, 11, 19, 16, 24, 20, 16, 19, 16, 18, 14. **(4 Points)**

5. WebStat 1.0 produces the following output for a least squares regression on the full (30 observations) “Cost of Victories” data set:

```
Regression results:
Parameter  Est          Std. Err.      Tstat      Pval
Intercept  11.43511840565116    1.8697244127040393    6.115937903978813    1.3404266043792745E-6
Slope      0.18388272396890235  0.0436344426765382244  2.14164606891476     2.3604482668249105E-4

se = 3.523698511037658
```

Write down the regression equation and indicate the predicted number of wins for a payroll of 50 (which relates to 50 Millions in the original data) based on this equation. Make clear what your variables stand for! (**3 Points**)

6. In class, we calculated $\hat{y} = 5.423 + 0.318x$ as our least squares regression line. So, is the result we got from WebStat 1.0 another bug in the software? Just state **yes** if you agree. If you disagree, provide an explanation for what has happened. (**3 Points**)
7. Clearly circle the point in the scatterplot in (5) above that will yield the biggest **absolute** (either positive or negative) residual. (**1.5 Points**)
8. After conducting some regression analysis in WebStat 1.0, are you fully satisfied? Aren't there many features we would like to see in a statistical package for regression analysis which are not supported in WebStat 1.0? Name **two** such features that would be very useful. (**3 Points**)